Grade 9 – Mathematics

Third Term Model Paper 4

Chalaka Perera | BSc (Hons) Computer Engineering, University of Peradeniya

Name / Index No.: ..... **Time: 3 hours**

- This paper consists of two parts: Part I and Part II.
- Part I: Answer **all** questions on this paper itself. Part A has 15 questions (2 marks each). Part B has 2 questions (10 marks each).
- Part II: Answer **five** questions only, on separate sheets. Each question carries 10 marks.
- Show all relevant steps of working. Write units where necessary. Use π = 22/7 where needed.

For examiner's use only	
PART I – A	/ 30
PART I – B	/ 20
PART II	/ 50
Total	**/ 100**

PART I

Part A – Answer all questions in the space provided. (2 marks each)

1. Write the next two terms of the number pattern 2, 6, 12, 20, 30, … (2)

Answer: .....

2. Which is greater, 11001_2 or 26? Show your working. (2)

Answer: .....

3. Find the value of ⅗ of 2½. (2)

Answer: .....

4. Increase Rs 250 by 20%. (2)

Answer: .....

5. Factorise: $6ab - 9ac$ (2)

Answer: .....

6. Find the value: $5^0 + 2^{-1}$ (2)

Answer: .....

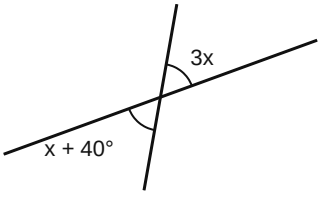
7. Simplify and give the answer in scientific notation: $(2 \times 10^3) \times (4 \times 10^2)$ (2)

Answer: .....

8. Solve: $3x/4 = 9$ (2)

Answer: .....

9. Two straight lines intersect as shown in the figure. (2)
Find the value of x.



Answer: .....

10. The angles of a triangle are x, x + 20° and x + 40°. Find the value of x. (2)

Answer: .....

11. The circumference of a circle is 88 cm. Find its diameter. ($\pi = 22/7$) (2)

Answer: .....

12. Find the length of a diagonal of a rectangle of length 8 cm and breadth 6 cm. (2)

Answer: .....

13. U = {1, 2, 3, …, 10} and A = {even numbers in U}. Write the elements of A′. (2)

Answer: .....

14. A letter is chosen at random from the letters of the word MATHEMATICS. Find the probability that it is M. (2)

Answer: .....

15. Find the sum of 1.5 l and 750 ml in litres. (2)

Answer: .....

Part B – Answer both questions on this paper. (10 marks each)

1.

A car uses 6 litres of petrol to travel 84 km. Assume that the amount of petrol used is directly proportional to the distance travelled.

(i) How many kilometres does the car travel with 1 litre? (2 marks)

(ii) How many litres are needed to travel 210 km? (2 marks)

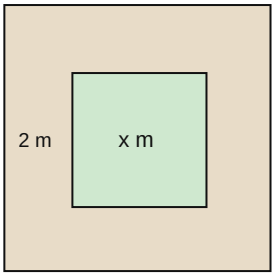
(iii) If 1 litre of petrol costs Rs 370, find the cost of the petrol for 210 km. (2 marks)

(iv) The price of petrol increases by 10%. Find the new price of 1 litre. (2 marks)

(v) Find the cost of the petrol for 210 km at the new price. (2 marks)

2.

A square lawn of side x m has a path 2 m wide all around it, as shown in the figure.



(i) Write the length of a side of the outer square in terms of x. (2 marks)

(ii) Write the area of the outer square in expanded form. (2 marks)

(iii) Show that the area of the path is $(8x + 16)$ m². (2 marks)

(iv) Factorise $8x + 16$. (2 marks)

(v) If the area of the path is 96 m², find the value of x. (2 marks)

PART II

Answer five questions only. (10 marks each)

1.

(a) A mobile phone bought for Rs 40 000 is sold at a loss of 15%. Find the selling price. (3 marks)

(b) The population of a village was 2500. It increased by 8% in one year and then decreased by 5% in the next year. Find the population at the end of the second year. (4 marks)

(c) Express 45 as a percentage of 180. (3 marks)

2.

(a) Expand and simplify: $(x + 4)(2x – 3)$ (2 marks)

(b) Factorise: (i) $x^2 + x – 20$ (ii) $2x^2 – 18$ (iii) $xy – 3x + 2y – 6$ (5 marks)

(c) Simplify: $3/(2a) – 1/(3a)$ (3 marks)

3.

(a) A father is three times as old as his son. In 10 years' time, the father will be twice as old as the son. Taking the son's present age as x years, form an equation and find their present ages. (4 marks)

(b) Solve the simultaneous equations: $2x + y = 11$, $x – 2y = –2$ (4 marks)

(c) The area of a trapezium is given by $A = ½h(a + b)$. Find h when $A = 60$, $a = 7$ and $b = 5$. (2 marks)

4.

An incomplete table of values for the function $y = ½x + 1$ is given below.

x	–4	–2	0	2	4
y	–1	…	1	…	3

(i) Fill in the blanks in the table. (1 marks)

(ii) Draw the graph of the function on graph paper. (3 marks)

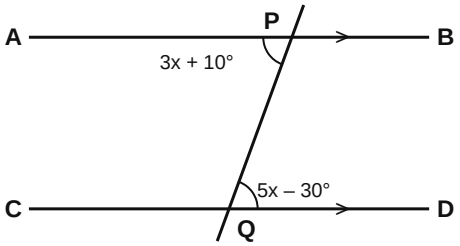
(iii) Write the gradient and the intercept of the graph. (2 marks)

(iv) Write the coordinates of the point where the graph meets the x-axis. (1 marks)

(v) Using the graph, find the value of x when $y = 2.5$. (2 marks)

(vi) Does the point (6, 4) lie on this line? Give a reason. (1 marks)

5.



(a) In the figure, AB ∥ CD. $A\hat{P}Q = 3x + 10°$ and $P\hat{Q}D = 5x – 30°$. (i) Find the value of x, giving reasons. (ii) Find the magnitude of $C\hat{Q}P$. (4 marks)

(b) The sum of the interior angles of a polygon is 1080°. (i) Find the number of sides. (3 marks)

(ii) If it is a regular polygon, find one interior angle.

(c) In triangle PQR, the exterior angle at R is 130° and $\hat{P} : \hat{Q} = 2 : 3$. Find $\hat{P}$ and $\hat{Q}$. (3 marks)

6.

(π = 22/7)

(a) A piece of wire 88 cm long is bent to form a circle.

(i) Find the radius and the area of the circle. (3 marks)

(ii) The same wire is bent to form a square. Find the area of the square. (2 marks)

(iii) Which shape encloses the larger area, and by how much? (1 marks)

(b) A rectangular field is 24 m long and 7 m wide. A boy walks from one corner to the opposite corner along the diagonal instead of along the two edges. How many metres does he save? (4 marks)

7.

The number of children in 20 families is given in the frequency table below. The frequency f is unknown.

No. of children	1	2	3	4	5
No. of families	3	7	f	4	2

(i) Find the value of f. (2 marks)

(ii) Find the total number of children in the 20 families. (2 marks)

(iii) Find the mean number of children per family. (2 marks)

(iv) Write the mode. (1 marks)

(v) Find the median. (3 marks)